

Estimated Annual Agricultural Pesticide Use Morrow County, Oregon 2014-2018

24 July 2026

Summary

Morrow County ranks **#1 of Oregon's 36 counties** by total estimated agricultural pesticide mass applied (EPest-high method, summed 2014-2018), making it one of OR's five highest pesticide-use counties. Morrow County, in the Columbia Basin/Columbia Plateau of north-central Oregon, is farmed under large-scale center-pivot irrigation around Boardman, producing potatoes, wheat, and alfalfa alongside significant food-processing activity.

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides in **Morrow County, Oregon** during 2018, as reported by [USGS Estimated Annual Agricultural Pesticide Use](#) data. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, and each compound's chemical class (abbreviated; see the key in Table 1 and the full classification, with a separate functional class column, in `data/pesticide_classification.csv`, shared with every chapter in this repository). Rankings and estimates are based on the *EPest-high method* described by [Thelin and Stone \(2013\)](#) and [Baker and Stone \(2015\)](#):

“EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD.” [-Pesticide National Synthesis Project webpage](#)

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. “detailed interpretation of where and how much use occurs within a county is not appropriate”).
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 -included windows once a complete 2019 release is available.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
CPX	Chlorophenoxy compound
DIP	Dipyridyl compound
DTC	Dithiocarbamates
INO	Inorganic compounds
MIC	Microbial
CB	N-methyl carbamates (AChE inhibitor)
OGM	Organo-metallic compound
OC	Organochlorine compound
OP	Organophosphorous compound

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)
(continued)

Code	Chemical class
OTH	Other
PYR	Pyrethrins/Pyrethroids
TC	Thiocarbamates
TRZ	Triazines

Table 2: Top 80 EPest-high Estimates, Morrow County, OR, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
1	METAM	DTC	1135539	858083	689997
2	DICHLOROPROPENE	OC	508383	426466	383259
3	BACILLUS AMYLOLIQUEFACIEN	MIC	309872	149088	149088
4	CHLOROPICRIN	OC	292588	104689	68752
5	METAM POTASSIUM	DTC	221006	169121	173608
6	SULFURIC ACID	INO	73678	148384	141297
7	GLYPHOSATE	OTH	56740	109472	108405
8	ETHOPROPHOS	OP	37874	24674	15148
9	BURKHOLDERIA SPP	MIC	21309	26474	17650
10	2,4-D	CPX	21234	28598	29275
11	PARAQUAT	DIP	20636	13422	9565
12	CHLOROTHALONIL	OTH	16638	16360	12303
13	EPTC	TC	16485	10422	9078
14	DCPA	OTH	15701	21215	19864
15	FLUFENACET	OTH	12388	8589	8193
16	ATRAZINE	TRZ	11545	15273	13245
17	METHOMYL	CB	9452	8006	8093
18	MANCOZEB	DTC	8934	26311	24500
19	MCPA	CPX	8137	11445	8093
20	METOLACHLOR & METOLACHLOR-S	OTH	7700	6771	7355
21	METOLACHLOR-S	OTH	7700	6771	7144
22	PETROLEUM OIL	OTH	7539	10365	9582
23	OXAMYL	CB	7212	4104	4002
24	COPPER HYDROXIDE	INO	7117	5562	5451
25	COPPER OXYCHLORIDE	INO	7114	4843	3353
26	TRIFLURALIN	OTH	5962	4291	5672
27	AZOXYSTROBIN	OTH	5080	3728	2922
28	DIURON	OTH	4648	13632	13917
29	DIMETHENAMID & DIMETHENAMID-P	OTH	4221	3701	3423
30	DIMETHENAMID-P	OTH	4221	3701	3423
31	DICAMBA	OTH	3925	3641	3152
32	FLUTOLANIL	OTH	3768	2187	1751
33	SULFUR	INO	3497	3260	3578
34	PENDIMETHALIN	OTH	3466	3928	4380
35	DIQUAT	DIP	2793	1608	1094
36	METRIBUZIN	TRZ	2613	2157	2791
37	FLUAZINAM	OTH	2456	1088	899
38	MALATHION	OP	2378	2993	3724
39	CLOPYRALID	OTH	2213	1660	1604
40	CHLORPYRIFOS	OP	2124	4674	4316
41	BROMOXYNIL	OTH	1922	2381	3124
42	TRICLOPYR	OTH	1887	1957	2228
43	DIMETHOATE	OP	1884	3321	2763
44	ETHOFUMESATE	OTH	1714	2692	2147
45	PROPICONAZOLE	OTH	1602	1692	1255
46	PYMETROZINE	OTH	1446	788	596

Table 2: Top 80 EPest-high Estimates, Morrow County, OR, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) (*continued*)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
47	MEFENOXAM	OTH	1422	1289	1063
48	PYRACLOSTROBIN	OTH	1402	1219	1273
49	BOSCALID	OTH	1389	1083	1181
50	FLUROXYPYR	OTH	1328	2373	1876
51	FLUOPYRAM	OTH	1273	526	378
52	CYHALOTHRIN-LAMBDA	PYR	1253	906	696
53	SPIROTETRAMAT	OTH	1192	884	725
54	PINOXADEN	OTH	1157	574	465
55	SAFLUFENACIL	OTH	1143	392	267
56	PYRIMETHANIL	OTH	1127	669	703
57	THIOPHANATE-METHYL	OTH	983	2197	1430
58	TOLFENPYRAD	OTH	868	868	868
59	SETHOXYDIM	OTH	828	581	439
60	PROPARGITE	OTH	807	1512	1845
61	BENZOVINDIFLUPYR	OTH	787	444	444
62	CYFLUTHRIN	PYR	787	625	541
63	FLONICAMID	OTH	736	466	340
64	CLETHODIM	OTH	669	967	776
65	BIFENTHRIN	PYR	623	310	256
66	IMIDACLOPRID	OTH	614	569	594
67	ALPHA CYPERMETHRIN	PYR	610	306	306
68	PYROXASULFONE	OTH	609	869	869
69	CYANTRANILIPROLE	OTH	605	405	373
70	2,4-DB	CPX	497	372	260
71	IPRODIONE	OTH	493	1560	1620
72	ESFENVALERATE	PYR	490	353	254
73	PENTHIOPYRAD	OTH	476	686	581
74	ACETOCHLOR	OTH	474	941	1024
75	IMAZAMOX	OTH	470	1030	941
76	TEBUCONAZOLE	OTH	385	876	861
77	FLUPYRADIFURONE	OTH	370	128	119
78	PROTHIOCONAZOLE	OTH	362	430	733
79	FLUXAPYROXAD	OTH	326	585	370
80	TERBACIL	OTH	313	156	144

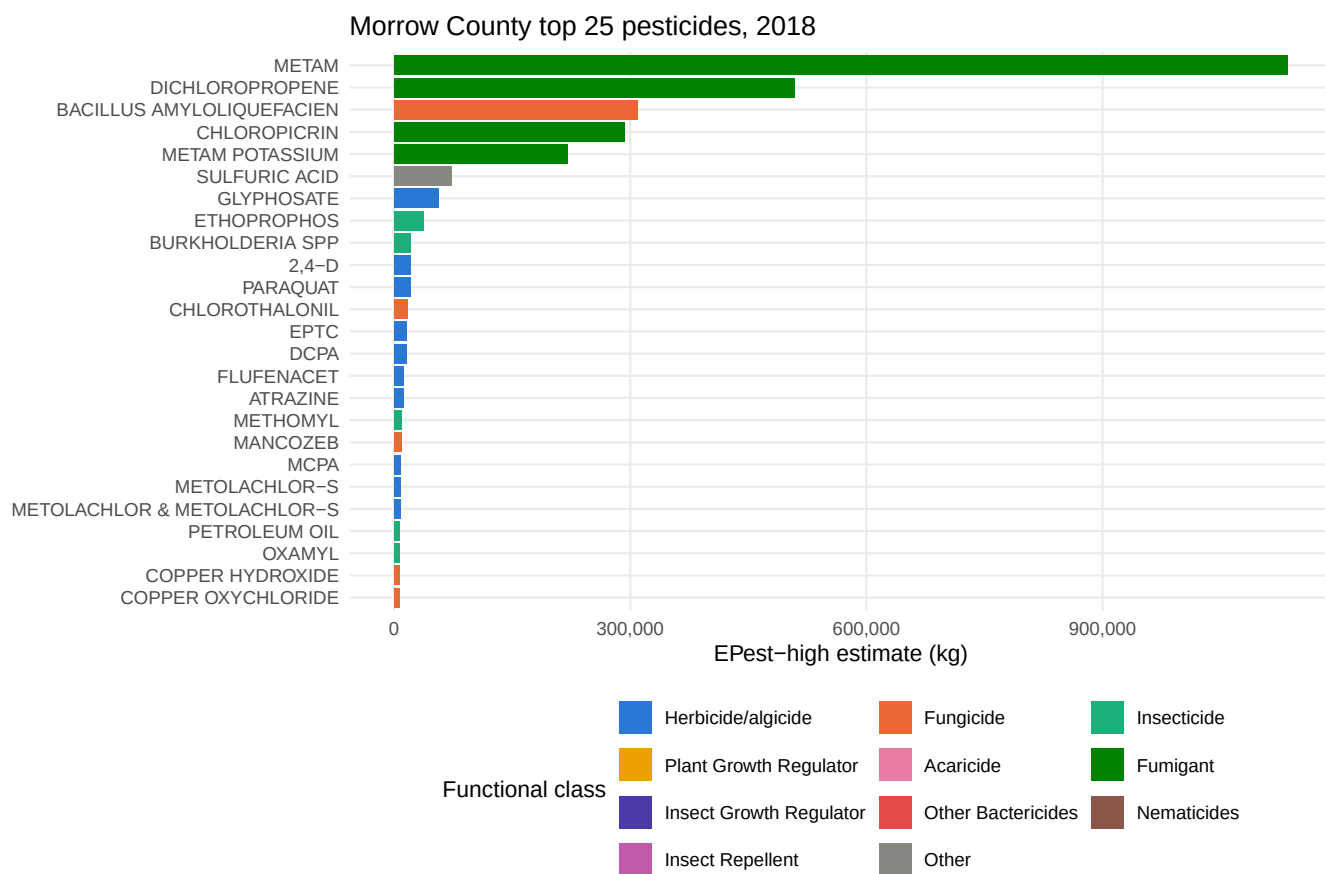


Figure 1: Top 25 compounds in Morrow County by E Pest-high mass applied, most recent year, colored by pesticide functional class.

Appendices

Table 3: Top 50 Morrow County Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	METAM	908597	1135539
2	DICHLOROPROPENE	508383	508383
3	BACILLUS AMYLOLIQUEFACIEN	309872	309872
4	CHLOROPICRIN	42088	292588
5	METAM POTASSIUM	0	221006
6	SULFURIC ACID	0	73678
7	GLYPHOSATE	55715	56740
8	ETHOPROPHOS	0	37874
9	BURKHOLDERIA SPP	21309	21309
10	2,4-D	20988	21234
11	PARAQUAT	5016	20636
12	CHLOROTHALONIL	12720	16638
13	EPTC	16485	16485
14	DCPA	0	15701
15	FLUFENACET	0	12388
16	ATRAZINE	4865	11545
17	METHOMYL	3230	9452
18	MANCOZEB	7226	8934
19	MCPA	1728	8137
20	METOLACHLOR & METOLACHLOR-S	6700	7700
21	METOLACHLOR-S	6700	7700
22	PETROLEUM OIL	7539	7539
23	OXAMYL	6473	7212
24	COPPER HYDROXIDE	7115	7117
25	COPPER OXYCHLORIDE	7114	7114
26	TRIFLURALIN	0	5962
27	AZOXYSTROBIN	0	5080
28	DIURON	63	4648
29	DIMETHENAMID & DIMETHENAMID-P	2663	4221
30	DIMETHENAMID-P	2663	4221
31	DICAMBA	1009	3925
32	FLUTOLANIL	3768	3768
33	SULFUR	3497	3497
34	PENDIMETHALIN	30	3466
35	DIQUAT	2793	2793
36	METRIBUZIN	1558	2613
37	FLUAZINAM	2456	2456
38	MALATHION	2	2378
39	CLOPYRALID	112	2213
40	CHLORPYRIFOS	1371	2124
41	BROMOXYNIL	1904	1922
42	TRICLOPYR	0	1887
43	DIMETHOATE	0	1884
44	ETHOFUMESATE	0	1714
45	PROPICONAZOLE	1315	1602
46	PYMETROZINE	0	1446
47	MEFENOXAM	0	1422
48	PYRACLOSTROBIN	36	1402
49	BOSCALID	5	1389
50	FLUROXYPYR	0	1328

Table 4: Top 50 Morrow County Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	METAM	649376	858083
2	DICHLOROPROPENE	426466	426466
3	METAM POTASSIUM	0	169121
4	BACILLUS AMYLOLIQUEFACIEN	149088	149088
5	SULFURIC ACID	0	148384
6	GLYPHOSATE	105887	109472
7	CHLOROPICRIN	15273	104689
8	2,4-D	27313	28598
9	BURKHOLDERIA SPP	21640	26474
10	MANCOZEB	25742	26311
11	ETHOPROPHOS	6	24674
12	DCPA	12606	21215
13	PROPAMOCARB HCL	0	21201
14	CHLOROTHALONIL	13006	16360
15	ATRAZINE	4898	15273
16	DIURON	72	13632
17	PARAQUAT	7240	13422
18	MCPA	751	11445
19	EPTC	9402	10422
20	PETROLEUM OIL	10365	10365
21	FLUFENACET	0	8589
22	METHOMYL	3879	8006
23	PHOSPHORIC ACID	4317	7523
24	METOLACHLOR & METOLACHLOR-S	6077	6771
25	METOLACHLOR-S	6077	6771
26	COPPER HYDROXIDE	5410	5562
27	QUINTOZENE	0	5490
28	COPPER OXYCHLORIDE	4772	4843
29	CHLORPYRIFOS	3160	4674
30	KAOLIN CLAY	1174	4647
31	TRIFLURALIN	19	4291
32	OXAMYL	2158	4104
33	PENDIMETHALIN	1195	3928
34	AZOXYSTROBIN	228	3728
35	DIMETHENAMID & DIMETHENAMID-P	1939	3701
36	DIMETHENAMID-P	1939	3701
37	DICAMBA	2095	3641
38	DIMETHOATE	7	3321
39	SULFUR	3260	3260
40	MALATHION	1420	2993
41	ETHOFUMESATE	818	2692
42	BROMOXYNIL	2356	2381
43	FLUROXYPYR	0	2373
44	THIOPHANATE-METHYL	848	2197
45	FLUTOLANIL	2144	2187
46	METRIBUZIN	1227	2157
47	TRICLOPYR	1	1957
48	PROPICONAZOLE	1425	1692
49	CLOPYRALID	190	1660
50	DIQUAT	1608	1608

Table 5: Top 50 Morrow County Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	METAM	510794	689997
2	DICHLOROPROPENE	375060	383259
3	METAM POTASSIUM	42224	173608
4	BACILLUS AMYLOLIQUEFACIEN	149088	149088
5	SULFURIC ACID	28763	141297
6	GLYPHOSATE	105469	108405
7	CHLOROPICRIN	14939	68752
8	2,4-D	28275	29275
9	MANCOZEB	24158	24500
10	PROPAMOCARB HCL	0	21201
11	DCPA	10784	19864
12	BURKHOLDERIA SPP	14427	17650
13	ETHOPROPHOS	60	15148
14	DIURON	67	13917
15	ATRAZINE	4646	13245
16	CHLOROTHALONIL	9368	12303
17	PETROLEUM OIL	9582	9582
18	PARAQUAT	5365	9565
19	EPTC	8237	9078
20	PHOSPHORIC ACID	7190	8793
21	FLUFENACET	0	8193
22	MCPA	527	8093
23	METHOMYL	3829	8093
24	METOLACHLOR & METOLACHLOR-S	5899	7355
25	METOLACHLOR-S	5711	7144
26	TRIFLURALIN	21	5672
27	COPPER HYDROXIDE	5317	5451
28	QUINTOZENE	1553	5213
29	PENDIMETHALIN	2166	4380
30	CHLORPYRIFOS	2713	4316
31	OXAMYL	2834	4002
32	MALATHION	2455	3724
33	KAOLIN CLAY	1586	3670
34	DIAZINON	3596	3600
35	SULFUR	3578	3578
36	DIMETHENAMID & DIMETHENAMID-P	1711	3423
37	DIMETHENAMID-P	1711	3423
38	COPPER OXYCHLORIDE	3294	3353
39	DICAMBA	1846	3152
40	BROMOXYNIL	2024	3124
41	ACEPHATE	2984	2984
42	METHYL BROMIDE	2977	2977
43	AZOXYSTROBIN	417	2922
44	METRIBUZIN	1046	2791
45	DIMETHOATE	39	2763
46	GLUFOSINATE	30	2496
47	TRICLOPYR	197	2228
48	ETHOFUMESATE	610	2147
49	FLUROXYPYR	41	1876
50	PROPARGITE	1727	1845